

# Dekai Li

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## BIOGRAPHY

Dekai Li is affiliated with the Department of Artificial Intelligence, School of Information and Communication Engineering, Hainan University. His research focuses on hyperspectral remote sensing, spectral-spatial representation learning, Mamba/state-space models, multimodal HSI-LiDAR fusion, contrastive learning, and limited-sample fine-grained classification, with applications to coastal and mangrove remote sensing.

## RESEARCH INTERESTS

Hyperspectral image classification; Mamba / state-space models; spectral-spatial fusion; HSI-LiDAR multimodal learning; Transformer and attention; contrastive learning; mangrove and coastal remote sensing.

## ACADEMIC AFFILIATION

**Hainan University** - Department of Artificial Intelligence, School of Information and Communication Engineering, Haikou, Hainan, China.

## PUBLICATIONS

**2026 - FACANet: Fourier Aware Contrastive Alignment Network for Fine-Grained Mangrove Classification with Limited Samples**

Dekai Li, Uzair Aslam Bhatti, Jiaxin Li, Songpeng Gong. *Expert Systems With Applications*, journal pre-proof. DOI: 10.1016/j.eswa.2026.134187.

**2026 - GS-MambaNet: A Decoupled Global-Spatial and Grouped-Spectral Mamba Network for Hyperspectral Image Classification**

Dekai Li, Uzair Aslam Bhatti, Jiaxin Li, Yonis Gulzar, Shahad Albannay, Ayoob Lone. *IEEE JSTARS*. DOI: 10.1109/JSTARS.2026.3702378.

**2026 - HyPyraMamba: A Pyramid Spectral Attention and Mamba-Based Architecture for Robust Hyperspectral Image Classification**

Dekai Li, Uzair Aslam Bhatti, Mengxing Huang, Lorenzo Bruzzone, Jiaxin Li. *IEEE TGRS*. DOI: 10.1109/TGRS.2025.3650350.

**2026 - HFMamba: An Advanced Coastal Paradigm for Hyperspectral Image Analysis in Consumer-Driven Agricultural Remote Sensing**

Dekai Li, Uzair Aslam Bhatti, Mengxing Huang, Junfeng Zhang, Mohammad Shabaz, Leila Jamel, Shulei Wu. *IEEE Transactions on Consumer Electronics*. DOI: 10.1109/TCE.2025.3645064.

**2025 - CSTFNet: A CNN and Dual Swin-Transformer Fusion Network for Remote Sensing Hyperspectral Data Fusion and Classification of Coastal Areas**

Dekai Li, Harold Neira-Molina, Mengxing Huang, Syam M.S., Yu Zhang, Zhang Junfeng, Uzair Aslam Bhatti, Muhammad Asif, Nadia Sarhan, Emad Mahrous Awwad. *IEEE JSTARS*. DOI: 10.1109/JSTARS.2025.3530935.

**2025 - DAFNet: A Dual-Branch Attention Fusion Network with Multi-Scale Residual Learning for Hyperspectral Image Classification**

Dekai Li, Uzair Aslam Bhatti, Otabek Safarbaev, Jumabayev Ikhlosbek Umidjon ugli, Mughair Asalm Bhatti, Hao Tang. *IEEE PRMVAI*. DOI: 10.1109/PRMVAI65741.2025.11108624.

## PATENTS

**2026 - Precise hyperspectral image classification method based on improved multiscale attention and Transformer network.** Patent No. ZL 2024 1 1967702.7; Publication No. CN 120047816 B; granted 23 Jun 2026.

**2025 - Precise hyperspectral image classification method based on CNN and Dual-Swin-Transformer fusion network.** Patent No. ZL 2024 1 1949888.3; Publication No. CN 119888329 B; granted 25 Nov 2025.

## ACADEMIC SERVICE

Reviewer for IEEE Transactions on Image Processing (TIP), IEEE Transactions on Geoscience and Remote Sensing (TGRS), International Journal of Applied Earth Observation and Geoinformation (JAG), IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JSTARS), Engineering Applications of Artificial Intelligence (EAAI), and Applied Soft Computing.

## OPEN-SOURCE & PROFILES

HyPyraMamba: [github.com/dekai-li/HyPyraMamba](https://github.com/dekai-li/HyPyraMamba) | FACANet: [github.com/dekai-li/FACANet](https://github.com/dekai-li/FACANet) | GitHub: [github.com/dekai-li](https://github.com/dekai-li) | Google Scholar: [scholar.google.com/citations?user=58hSyu0AAAAJ](https://scholar.google.com/citations?user=58hSyu0AAAAJ)